# The PKM isogeny-tower auxiliary curve at the NIST P-192 prime

A 2-Sylow halving search: cost per draw, success probability, and a supersingular shortcut

2026-10-15 · branch `claude/isogeny-tower-p192` · protocol `fc6317e51` (committed before data) · run `20261015T2010Z`

## Summary


- **Built.** `two_sylow_search.rs` and the `pkm-aux-search` binary. Each draw is a curve with full rational 2-torsion over $p = 2^{192} - 2^{64} - 1$. Its 2-Sylow is $\mathbb{Z}/2 \times \mathbb{Z}/2^b$, and x-only point halving computes $b$ with no point counting. A point of order $2^t$ exists iff $b \geq t$. Found points are certified over `BigUint` and accepted by the unmodified checker `backlog_check_aux2t`.
- **Per draw (measured).** On the $X_1(2, 8)$ family (`z2xz8`, $b \geq 3$ forced), the median is **10.06 µs**, against 30.83 µs for the baseline (0.326×). On `z2xz2` it is 13.89 µs (0.561× of 24.74). All four pre-registered speed claims pass.
- **Success probability (measured).** $2^{28}$ `z2xz8` draws and $2^{25}$ `z2xz2` draws match $P(b \geq t) = 2^{3-t}$ and $3 \cdot 2^{-t}$ inside 99.9% Poisson bands at every tested $t$. The deepest ordinary curve has $b = 38$.
- **$t = 48$ (derived).** $2^{45}$ draws × 10.06 µs = **98,352 core-hours ≈ 11.2 core-years**.
- **Shortcut (measured).** $y^2 = x^3 + x$ is supersingular over this prime, with $\#E = p + 1$ and $v_2(p + 1) = 64$. One draw gives a point of order $2^{48}$ (or $2^{64}$), and the checker accepts it. Whether PKM's relation phase accepts a supersingular $E'$ is untested.


## Method

For $p \equiv 3 \pmod 4$ and $E : y^2 = (x - e_1)(x - e_2)(x - e_3)$, the Weil pairing rules out $\mathbb{Z}/4 \times \mathbb{Z}/4$, so $E(\mathbb{F}_p)[2^\infty] \cong \mathbb{Z}/2 \times \mathbb{Z}/2^b$. A point $P$ lies in $2E$ iff all $x - e_i$ are squares. The halves of $\pm P$ have $x_H = x + r_1 r_2 + r_1 r_3 + r_2 r_3$ with $r_i = \pm\sqrt{x - e_i}$, $x_H - e_i = (r_i + r_j)(r_i + r_k)$ and $y_H = \pm(r_1 + r_2)(r_1 + r_3)(r_2 + r_3)$. Negating $r_n$ gives $H + T_n$. The descent map $\delta : E/2E \to \{\pm 1\}^2$ reduces the question "can some half be halved again?" to one symbol $\chi(x_H - e_k)$. The `fast` arm reads that symbol from the exponentiation the next level needs anyway, $i_k = (x_H - e_k)^{(p-3)/4}$ (189 squarings). A second exponentiation $u$ decides between $H$ and $H + T_n$ and yields the next roots without inversions.

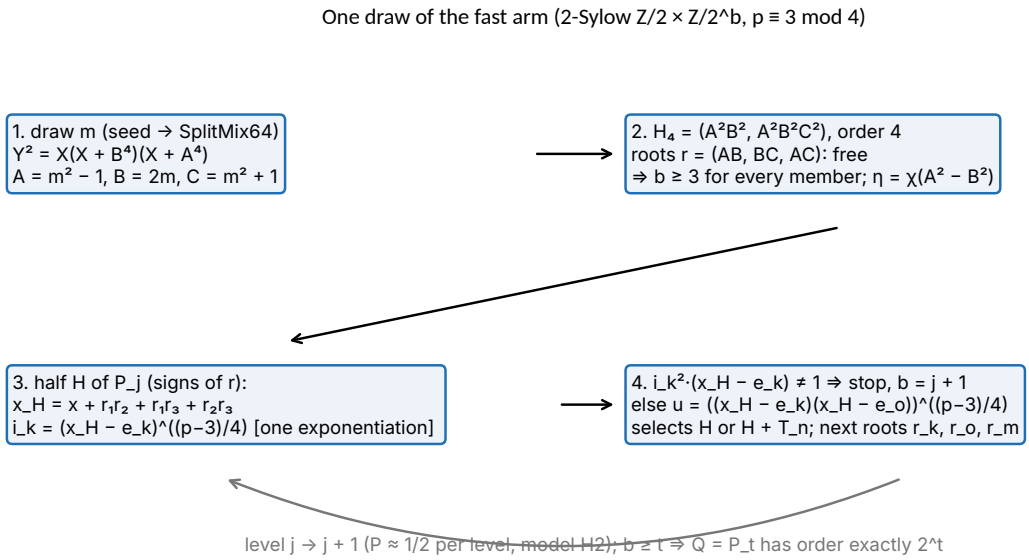



Figure 1: One draw of the `fast` arm. Measured per `z2xz8` draw (C4 counters): 4.0 exponentiations, 1.0 Legendre symbol, 1.0 halving.

## Cost per draw (C1, C2)

| family | arm | median µs | min | max | Legendre/draw | exps/draw |
|---|---|---|---|---|---|---|
| z2xz8 | baseline | 30.83 | 29.30 | 32.05 | 8.49 | 6.00 |
| z2xz8 | fast-chi | 12.48 | 12.15 | 13.30 | 4.00 | 2.00 |
| z2xz8 | **fast** | **10.06** | 9.78 | 10.19 | 1.00 | 4.00 |
| z2xz2 | baseline | 24.74 | 24.00 | 25.72 | 9.15 | 4.49 |
| z2xz2 | fast-chi | 16.81 | 16.70 | 17.63 | 5.24 | 2.99 |
| z2xz2 | fast | 13.89 | 13.52 | 14.96 | 3.00 | 3.93 |

Table 1: C1: 65,536 draws × 7 interleaved repetitions, one core (`taskset -c 2`) of a shared 4-core host (load 2.3–3.5). Pre-registered: S1 fast/baseline 0.326 ≤ 0.40 ✓; S2 fast/fast-chi 0.806 ≤ 0.85 ✓; S3 10.06 ≤ 12 µs ✓; S4 (`z2xz2`) 0.561 ≤ 0.65 ✓. No min–max spreads overlap.

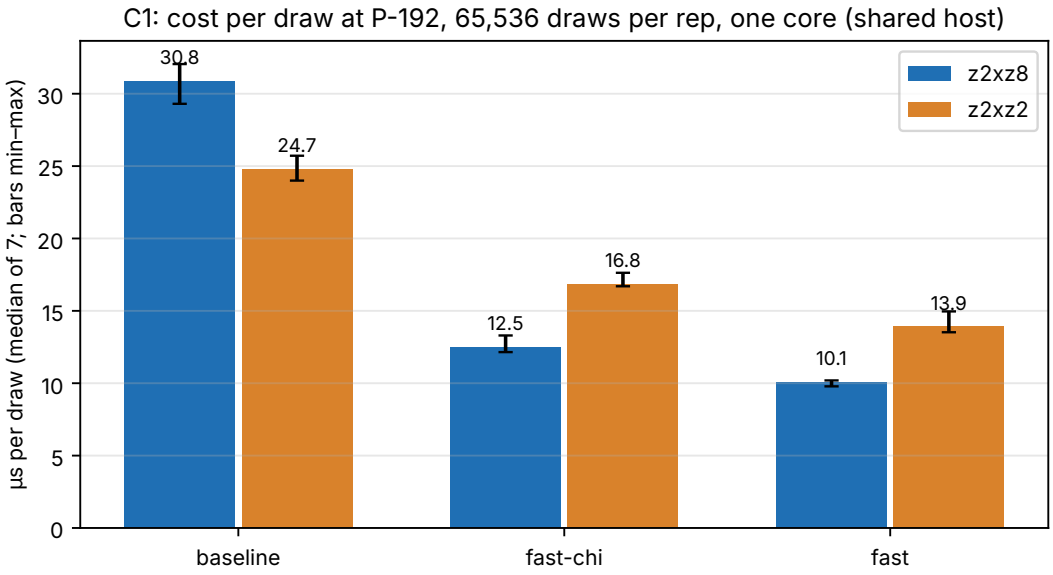



Figure 2: C1 medians with min–max bars.

C2 (ns per call, median of 11): mul 10.6; dedicated squaring 5.1; $(p-3)/4$ chain 1,808; `sqrt` 2,185; binary Jacobi 1,793; posdivsteps Jacobi 1,952; inversion 2,153; drawing $m$ 146. **Negative result:** the exact posdivsteps Jacobi (libsecp256k1-style) is not faster than the binary one at 192 bits. Interleaving two exponentiation chains was slower and was dropped before the protocol.

## Success probability (C3, C4)

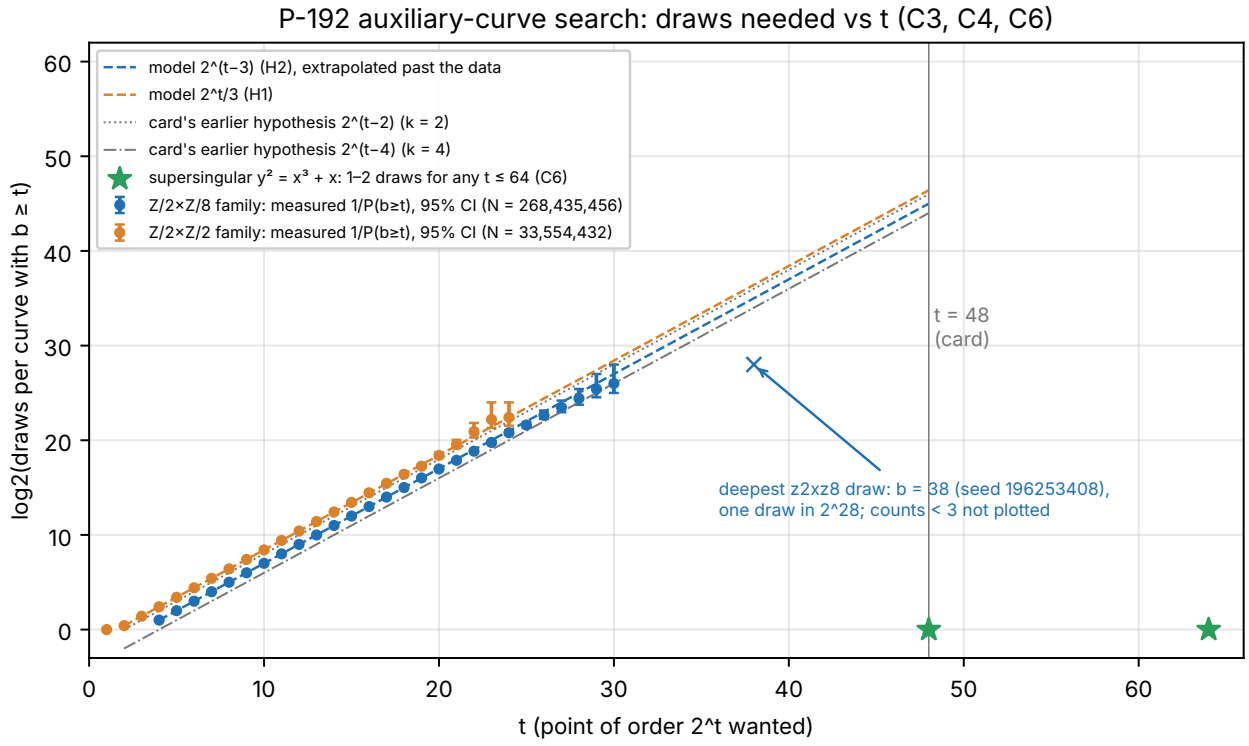



Figure 3: Measured draws per success $1/\hat{P}(b \geq t)$ with 95% intervals (C3: $N = 2^{25}$ `z2xz2`; C4: $N = 2^{28}$ `z2xz8`). Dashed: the pre-registered models, extrapolated to $t = 48$. Dotted: the card's earlier hypotheses. Stars: the supersingular curve (C6).

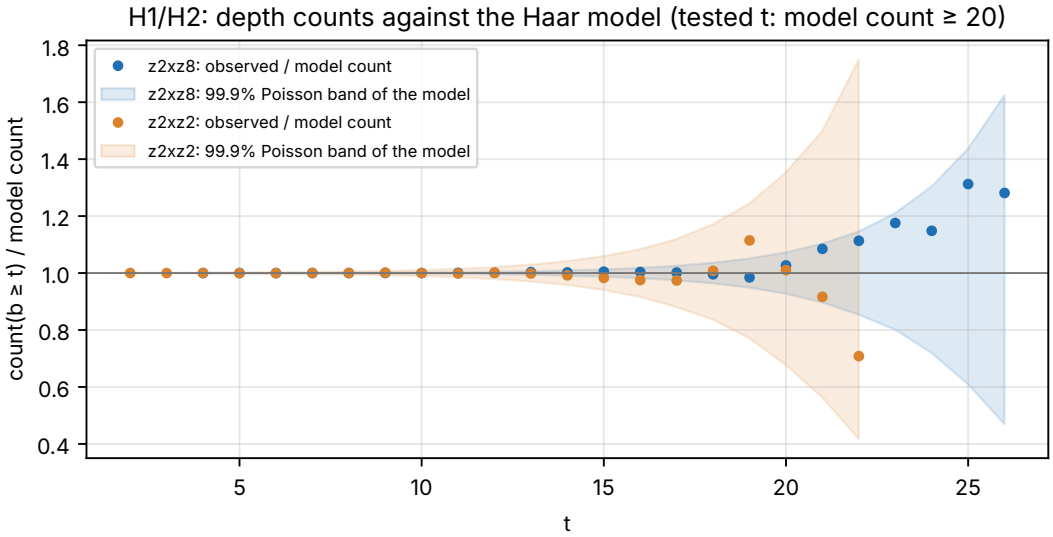



Figure 4: Observed / model count of draws with $b \geq t$, with the model's 99.9% Poisson band; only $t$ with model count ≥ 20 is tested. H1 and H2: consistent at every tested $t$. For `z2xz8` the ratio drifts above 1 for $t \geq 21$ (up to 1.31 at $t = 25$) while staying inside its band. The counts are cumulative, so neighbouring ratios are not independent. This is recorded as an observation.

| $t$ | count $b \geq t$ | model | ratio | mean draws to success | model |
|---|---|---|---|---|---|
| 16 | 32,894 | 32,768 | 1.004 | 8,161 ± 45 | 8,192 |
| 20 | 2,103 | 2,048 | 1.027 | 127,621 ± 2,772 | 131,072 |
| 24 | 147 | 128 | 1.148 | 1,822,411 ± 157,208 | 2,097,152 |

| | | | | | |
|---|---|---|---|---|---|
| 28 | 12 | 8 | 1.5 | (22.6 ± 5.2)·10⁶ | 33.6·10⁶ |

`z2xz2` (C3) gives $P(b = 1)$: 8,387,239 observed against 8,388,608 predicted. The deepest `z2xz8` draw has $b = 38$ (seed 196253408), a 1-in-128 upper-tail event under H2.

## Curves, checker and tower (C5–C7)

`backlog_check_aux2t --expect-t t` accepts the ordinary artifacts at $t = 16, 24, 28, 38$ and rejects $t + 1$. It also accepts the supersingular artifacts at $t = 48$ and 64. The tower $V = x(R' + \langle Q \rangle)$, built at P-192 on the $b = 38$ curve ($T = 8, 12, 16$) and on the supersingular curve ($T = 8, 16$), has $|V| = 2^T$ distinct abscissae. $L(x) = c$ holds on all of $V$ in the field backend and again over `BigUint`. None of 4096 random abscissae outside $V$ hits $c$. About half of $V$ lifts to P-192 (32,609 of 65,536 at $T = 16$).

## The $t = 48$ cost, and what it does not decide

$2^{45}$ `z2xz8` draws × 10.06 µs = 98,352 core-hours (95,617–99,624 over the C1 spread), about 12 core-hours per $2^{32}$-draw shard. `z2xz2` would need 362,005. The card's earlier 39,094 core-hours assumed $2^{46}$ draws at 2 µs. $\mathbb{Z}/2 \times \mathbb{Z}/8$ buys 3 bits, not 4, because $v_2(p - 1) = 1$ caps the non-cyclic part at $\mathbb{Z}/2$. The figure is derived: it extrapolates H2 seventeen bits past the deepest tested $t$ on one prime.

The supersingular curve removes the search for P-192, and for any prime with large $v_2(p + 1)$, if a supersingular $E'$ is admissible for PKM. This study does not test that, nor whether a long tower helps PKM at all (encyclopedia §8 lead 1).

Data: `data/` (C1–C7 raw JSON), `derived.json` (from `examples/two_sylow_study_stats.rs`), figures from `figures/plot.py` (layout only). Code: `src/cryptanalysis/two_sylow_search.rs`, `src/bin/pkm_aux_search.rs`, `examples/{two_sylow_bench,p192_isogeny_tower,two_sylow_study_stats}.rs`.